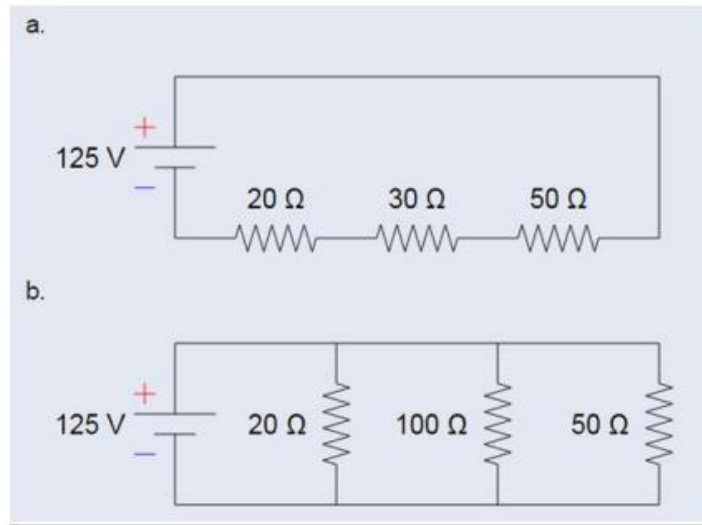


PART A



For each of these circuits:

1. Calculate the total resistance in ohms,
2. Calculate the total current from the power supply,
3. Calculate the current through each resistor
4. Calculate the Voltage drop through each resistor
5. Calculate the Power dissipated through each resistor (*use Watt's Law: Power = Voltage * Current*)

1.

a) $R_1 + R_2 + R_3 = R_{\text{total}}$ – In series

$$20\ \Omega + 30\ \Omega + 50\ \Omega = 100\ \Omega$$

b) $1/R_1 + 1/R_2 + 1/R_3 = 1/R_{\text{total}}$ – In parallel

$$1/20\ \Omega + 1/100\ \Omega + 1/50\ \Omega = 12.5\ \Omega$$

2.

a)

$$I = V/R$$

$$125/100 = 1.25\text{A}$$

b)

$$125/12.5 = 10A$$

3.

a)

In series, current is the same throughout

$$I_1 = I_2 = I_3 = 1.25 A$$

b)

$$I = V/R$$

$$125/20 = 6.25A$$

$$125/100 = 1.25A$$

$$125/50 = 2.5A$$

4.

a)

$$V = IR$$

$$1.25 * 20 = 25V$$

$$1.25 * 30 = 37.5V$$

$$1.25 * 50 = 62.5V$$

b)

$$V = IR$$

In parallel the supply of voltage is the same

$$V=125V$$

5.

a)

$$P = V * I$$

$$25 * 1.25 = 31.25W$$

$$37.5 * 1.25 = 46.875W$$

$$62.5 * 1.25 = 78.125W$$

$$\textbf{Total Power} = \textbf{156.25W}$$

b)

$$\textbf{P} = \textbf{V} * \textbf{I}$$

$$125 * 6.25 = 781.25W$$

$$125 * 1.25 = 156.25W$$

$$125 * 2.5 = 312.5W$$

$$\textbf{Total Power} = \textbf{1250W}$$